Exercise 1. Let $E = \{1/n : n \in \mathbb{N}\}$. Prove that the function

$$f(x) = \begin{cases} 1 & x \in E \\ 0 & \text{otherwise} \end{cases}$$

is integrable on $[0, 1]$. What is the value of $\int_0^1 f(x)\,dx$?

Proof. Let $a, b \in [0, 1]$ with $a < b$. Then there is a real number $\xi \in [a, b] \setminus \mathbb{Q}$. Thus, we have $\inf_{x\in[a,b]} f(x) = 0$, and it follows that $L(f, P) = 0$ if P is any partition of $[0, 1]$.

Given $\varepsilon > 0$, use the Archimedean Principle to choose an $n \in \mathbb{N}$ such that $2 < n\varepsilon$ with $n > 1$. Then define

$$E_k := \left\{ |x_k - y_k| : \frac{1}{n-k+1} < x_k < y_k < \frac{1}{n-k} \right\}$$

for $k = 1, 2, \ldots, n-1$. Since $\sup E_k = \frac{1}{n-k} - \frac{1}{n-k+1}$ for $1 \le k \le n-1$, there exist points $x_k, y_k \in \left(\frac{1}{n-k+1}, \frac{1}{n-k}\right)$ such that

$$\frac{1}{n-k} - \frac{1}{n-k+1} - \frac{\varepsilon}{2(n-1)} < |x_k - y_k|$$

with $x_k < y_k$ for each $1 \le k \le n-1$.

Now, set $P := \left\{0, \frac{1}{n}, x_1, y_1, x_2, y_2, \ldots, x_{n-1}, y_{n-1}, 1\right\}$, then P is a partition of $[0, 1]$. Thus,

$$\begin{aligned} U(f, P) - L(f, P) &= 1 - \sum_{k=1}^{n-1} |x_k - y_k| \\ &< 1 - \sum_{k=1}^{n-1} \left(\frac{1}{n-k} - \frac{1}{n-k+1} - \frac{\varepsilon}{2(n-1)} \right) \\ &= 1 - \left(1 - \frac{1}{n} - \frac{\varepsilon}{2} \right) \\ &= \frac{1}{n} + \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Therefore, f is integrable on $[0, 1]$. Hence, we obtain $\int_0^1 f(x)\,dx = \sup L(f, P) = 0$. $\square$